

TINNITUS THERAPY SUITE

User Manual

Version 2026.05.2

Open-source, browser-based sound therapy tools for tinnitus research and experimentation.

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SAFETY & LEGAL DISCLAIMER

DISCLAIMER

Official Web Access: tinnitus.trahreg.com

This tinnitus therapy suite is an experimental, educational, and informational tool.

It is not a medical device and is not intended to diagnose, treat, cure, or prevent any disease or health condition.

No medical claims are made or implied. The sound tools provided here are based on general concepts found in tinnitus research, but they are not a substitute for professional evaluation or treatment by a licensed clinician, audiologist, or healthcare provider.

AI-GENERATED CONTENT:

Certain features (CBT Reframing, SOS Spike Support, Sound Recipes, Pattern Identification, and Clinical Summaries) utilize Artificial Intelligence (Google Gemini). AI can "hallucinate" or provide inaccurate information. All AI-generated content is for educational exploration only. Do not make medical decisions based on AI suggestions without consulting a professional.

Use of this software is voluntary and at your own risk. Always keep listening volume at a safe and comfortable level. Stop using the tool immediately if you experience pain, discomfort, dizziness, or worsening symptoms.

If you have concerns about tinnitus, hearing loss, hyperacusis, or any other medical condition, consult a qualified healthcare professional.

This software is provided "AS IS," without warranty of any kind, express or implied. The authors and contributors are not liable for any damages, losses, or adverse effects arising from the use of this software.

By using this tool, you acknowledge that you have read and understood this disclaimer.

Official Web Access: <https://tinnitus.trahreg.com>

1. INTRODUCTION & SUCCESS PROTOCOLS

The Trahreg Tinnitus Therapy Suite is a research-oriented platform designed to help users transition through the stages of tinnitus habituation. Habituation is the process where the brain learns to ignore the tinnitus signal as a neutral, non-threatening sound.

The Five Golden Rules

Rule 1: The Mixing Point. Therapy volume must be set so the sound and your tinnitus "mix" like two colors of paint. Total masking (hiding the tinnitus completely) prevents habituation; your brain must still perceive the signal to learn it is neutral.

Rule 2: Passive Listening. Treat therapy like background "wallpaper." Do not focus on the sound; instead, engage in work, reading, or relaxation while the engine runs.

Rule 3: Consistency. Neuroplasticity takes time. Aim for 30–60 minutes of focused therapy daily for at least 3 months.

Rule 4: AI Support. If you experience a "spike" in distress, use the AI-powered **CBT Assistant** for immediate reframing and coping strategies rather than simply increasing the volume.

Rule 5: Clinical Visibility. Regularly export your **Clinical PDF Reports** to share objective progress (THI and MML trends) with your audiologist.

Rule 6: Audio Integrity. You must disable OS-level enhancements like Windows "Sonic," Dolby Atmos, or "Bass Boost," as these distort the calibrated therapy filters.

Protocol for Success (Phases)

Phase 1: Validation. Run **System Validation** weekly. This ensures your headphones are in-phase and the browser audio engine is delivering accurate signals.

Phase 2: Pitch Matching. Use the **Notch Finder**. Accuracy is critical; if your match is off by more than 5%, Notch and CR therapies lose clinical effectiveness.

Phase 3: Calibration. Set your therapy volume to the **Mixing Point** (slightly *lower* than your tinnitus). Total masking prevents the brain from habituating.

Phase 4: Active Habituation. Use the session timer and engage in non-auditory tasks. The goal is for the sound to fade into the background.

Phase 5: Data Correlation. Review "AI Insights" weekly to identify if sleep, stress, or environment are triggering spikes.

Phase 6: Monthly Review. Every 30 days, generate a Clinical PDF. Look for a 7-point drop in THI scores as a marker of success.

Using the Help Systems

The Guide: Accessible via the "Clinical Guide" button, this panel provides immediate access to the Golden Rules and Success Protocols from any page.

Interactive Tutorials: Click the "Tutorial" button on any therapy page for a step-by-step narrated walkthrough.

Professional Narrator: High-quality audio guidance is available for all setup steps. You can adjust the narrator's speed and volume in the tutorial menu.

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AI Assistant: Integrated into the CBT and Insight modules for real-time reframing and sound recipe suggestions.

2. DECORRELATED NOISE THERAPY

Purpose: Tinnitus is often associated with pathological neural synchrony. Decorrelation presents independent, unrelated noise signals to each ear, preventing the brain from "merging" the sound into a central image. This aims to disrupt the synchronized firing patterns in the auditory cortex.

Step-by-Step Tutorial

Hardware: Connect high-quality headphones. True decorrelation requires two discrete signals.

Sound Choice: Select a noise color (**Pink** is recommended for long-term comfort).

Calibration: Adjust the **Decorrelation** slider until the sound feels "wide" and fills your head, rather than being a single point.

Volume: Set to the **Mixing Point**.

Field Guide

Decorrelation Slider: Controls the independence of the L/R channels. 0% is identical (Mono); 100% is fully unrelated noise.

Noise Color:

Pink: Equal energy per octave; mirrors natural sound distribution.

White: Equal energy per frequency; effective for aggressive masking.

Brown: Heavy low-frequency emphasis; best for low-pitched "roaring" tinnitus.

EQ Sliders: Manually boost frequencies (2kHz to 12kHz) where you have known hearing loss to reduce listening effort.

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3. NOTCH THERAPY

Purpose: Based on **Tailored Notched Music Training (TMNMT)**, this therapy removes sound energy from a narrow band centered on your tinnitus pitch. This stimulates **Lateral Inhibition**: the stimulated neurons *surrounding* the notch actively suppress the hyperactive, tinnitus-producing neurons *inside* the notch. (Ref: *Okamoto et al., 2010*)

Step-by-Step Tutorial

Pitch Find: Use the **Notch Finder** to determine your center tinnitus frequency.

Frequency Entry: Input that frequency into the **Notch Center** field.

Width: Ensure **Notch Width** is set to **1.0 octaves** (the clinical standard).

Mixing: Set the volume so you can still perceive your tinnitus "inside" the silent notch in the noise.

Field Guide

Notch Center (Hz): The frequency of your tinnitus. Every slider movement here shifts the "silence" in the sound.

Notch Width (Octaves): The size of the frequency "hole." 1.0 is standard; smaller widths require higher pitch-matching precision.

Source Selector: Choose between **Internal Noise**, a **Custom MP3**, or **Live Audio** (e.g., Spotify processed through the notch).

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4. DUAL-STIMULUS NEUROMODULATION

Purpose: Inspired by bimodal devices like **Lenire**, this experimental module pairs structured auditory patterns (carrier tones, modulated noise, and tone bursts) with tactile feedback (haptics). By stimulating both the auditory and somatosensory systems simultaneously, the brain is encouraged to re-prioritize external signals over the internal tinnitus signal. (*Ref: TENT-A Trial, 2020*)

Step-by-Step Tutorial

Device: Best used on a smartphone held in the hand.

Activation: Enable **Tactile Haptic Pulse**.

Sensitivity: Adjust the **Trigger Sensitivity** until the phone vibrates *only* when you hear a distinct audio burst.

Rhythm: Use the **Pulse Rate** slider to match the visual pacer to your resting breathing rate (usually 5-6 BPM).

Field Guide

Carrier Freq (Hz): Sets the pitch of the tone bursts. Ideally set near your tinnitus frequency or a complementary low tone.

Noise Mod (Hz): The speed at which the background noise "shimmers" (amplitude modulation).

Burst Rate (Hz): How many audio pulses occur per second.

Tone/Noise Mix: Slides between pure bursts (1.0) and pure modulated noise (0.0). A 50/50 mix is standard.

Haptic Strength: Controls how forcefully the phone vibrates during a trigger.

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5. BROADBAND SOUND THERAPY

Purpose: The foundational tool for **Tinnitus Retraining Therapy (TRT)**. It provides constant, neutral sound enrichment to reduce the "contrast" between the tinnitus signal and a silent environment, preventing the brain's internal gain from increasing.

Step-by-Step Tutorial

Selection: Choose a sound you find neutral (**Pink Noise**) or pleasant (**Rain/Ocean**).

Calibration: Use the **Mixing Point** rule. The sound should be audible but *not* hide the tinnitus.

Duration: Use for 24/7 enrichment at low levels, or set a 60-minute timer for a focused session.

Field Guide

Noise Selection: Buttons for White, Pink, Brown, Blue, and Violet. Pink is the research standard for habituation.

L/R Balance: Moves the sound toward your "tinnitus side." Important if your tinnitus is louder in one ear.

Sleep Fade: When checked, the volume will slowly ramp down to zero over 60 seconds at the end of your timer to prevent an autonomic "rebound" spike.

4-7-8 Pacer: A visual and auditory breathing guide. Inhale for 4s, Hold for 7s, Exhale for 8s. Use during high distress spikes.

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6. ACOUSTIC COORDINATED RESET (CR)

Purpose: Developed by **Prof. Peter Tass**, this neuromodulation protocol uses pseudo-random sequences of four tones mathematically calculated relative to your tinnitus pitch. The goal is to desynchronize the pathological neural loops responsible for tonal tinnitus. (*Ref: Tass et al., 2012*)

Step-by-Step Tutorial

Calibration: Use the **Pitch Bracketing** tool to find your exact tinnitus frequency.

Calculation: Enter this frequency into the **Base Frequency** field and click "Calculate."

Balancing: This is critical. Play the sequence and adjust the **Bal 1-4** sliders until all four tones sound *exactly as loud as each other* to your ears.

Protocol: Follow the "3-on/2-off" pattern. 60-minute sessions are recommended.

Field Guide

Tone Grid: Displays the active therapeutic frequencies. A highlighted box indicates which tone is currently firing.

Bal Sliders (1-4): Adjusts individual tone volumes. Use these to compensate for hearing loss at specific frequencies.

Cycle Dots: A row of 5 dots. 3 solid dots (Active cycles) and 2 dashed dots (Silence cycles). This ensures the brain does not "adapt" to the sequence.

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7. BINAURAL BEATS & WELLNESS

Purpose: Utilizes the **Frequency Following Response (FFR)**. By playing two slightly different frequencies in each ear, the brain "perceives" a third, internal beat frequency. This can be used to induce Alpha, Theta (relaxation), or Delta (deep sleep) states, helping to down-regulate a stressed nervous system.

Field Guide

Stereo Separation: You MUST use headphones. Without discrete L/R separation, the binaural effect is lost.

Delta (1-4Hz): Induces deep, restorative sleep.

Theta (4-8Hz): Induces meditative, relaxed states; ideal for anxiety management.

Alpha (8-13Hz): Promotes calm focus.

8. TINNITUS HANDICAP INVENTORY (THI)

Purpose: A 25-item clinically validated tool that measures the functional, emotional, and catastrophic impact of tinnitus on your life. It categorizes severity into Grade 1 (Slight) through Grade 5 (Catastrophic).

Usage: Take the assessment every 30 days. Your progress is measured by a reduction in this score, not necessarily a change in the volume of the sound. **Page 8**

9. COGNITIVE RESTRUCTURING

Purpose: Tinnitus distress is often driven by the **Limbic System** (the brain's emotion center). This tool uses interactive **Thought Records** to help you identify negative "Automatic Thoughts" and reframe them into objective "Balanced Thoughts," facilitating habituation.

How to Use

Log: Record a situation where tinnitus was distressing.

Identify: Write down the thought you had (e.g., "This will never stop").

AI Assist: Use the **Balanced Thought** button to generate a clinical TRT reframing suggestion.

10. RELAXATION TRAINING

Purpose: Tinnitus and physiological stress exist in a feedback loop. This module provides **Progressive Muscle Relaxation (PMR)** guides and the **4-7-8 Breathing Pacer** to de-escalate the autonomic nervous system.

11. PERSONALIZED RECOMMENDATIONS

Purpose: An algorithmic tool that analyzes your THI scores and MML levels to suggest the most effective therapy for your specific grade of tinnitus. For example, high distress levels will prioritize CBT and Relaxation, while lower distress with tonal tinnitus will prioritize Notch therapy.

Mental Health Spike Protocol

If a tinnitus "spike" causes high emotional distress, clinicians recommend switching focus from Sound Therapy to **CBT & Wellness**. Managing the limbic system's reaction—getting your body out of "fight or flight"—is **Page 9** important than the acoustic signal during a spike.

12. NOTCH FINDER

Purpose: Accurate pitch matching is the foundation of Notch and CR therapies. The **Notch Finder** identifies your "Center Frequency"—the specific pitch of your tinnitus.

How to Match Your Pitch

Environment: Sit in a quiet room with headphones.

Search: Move the **Frequency Slider** slowly until the external tone matches your internal sound.

Octave Confusion Check: This is the most common error. Once you find a match, click the **Double (x2)** and **Half (/2)** buttons. Often, a lower frequency is the "true" match.

Two-Tone Comparison: If you are stuck, use the **Two-Tone** module to compare Tone A and Tone B to "bracket" the sound.

Field Guide

Frequency (Hz): The current test pitch. Can be entered manually for precision.

Auto-Sweep Speed: Sets the rate at which the tone automatically climbs the frequency range. Many users find it easier to identify their pitch while the sound is moving.

Save Frequency: Saves the match globally so it is automatically applied to Notch and CR therapy modules.

13. TWO-TONE COMPARISON

Purpose: Diagnostic aid to avoid octave errors. Allows A/B comparison between two distinct frequencies.

14. FREQUENCY SWEEP

Purpose: A 10-second logarithmic sweep from 20Hz to 20,000Hz. Useful for identifying "dead zones" in hearing or specific frequencies that trigger a "reactive" increase in tinnitus volume.

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15. TMC, LG, AND RI TESTING

Tinnitus Masking Curve (TMC): Maps your **Minimum Masking Level (MML)** across the spectrum. A "steep" curve calculates a high **Q-factor**, indicating tonal tinnitus, while a flat curve suggests broadband noise-like tinnitus.

16. LOUDNESS GROWTH (LG)

Purpose: Measures how your subjective perception of loudness grows relative to objective volume increments. This detects **Hyperacusis** (sound sensitivity) or **Recruitment**. *Tutorial: Play the tone and rate its loudness from "Very Soft" to "Uncomfortable."*

17. SUPPRESSION TEST (RI)

Residual Inhibition (RI): Measures the temporary silence or volume reduction after 60 seconds of high-level sound exposure.

Step 1: Play noise at a level that completely hides your tinnitus for 60 seconds.

Step 2: The noise stops. Use the stopwatch to time how many seconds it takes for your tinnitus to return to normal.

Insight: Increasing RI duration over months is a positive indicator of auditory system responsiveness.

18. HEARING RANGE TEST

Purpose: Tinnitus is often a "compensation" for auditory deprivation. Use this tool to explore your auditory thresholds across the spectrum (20Hz to 20kHz) and identify shifts in hearing.

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19. AI & SUPPORT TOOLS

AI-Powered Features (Gemini): Leverages client-side AI (via your own API key) to provide therapeutic support. All data is anonymized before processing.

"Spike" De-escalator: Provides immediate CBT reframing during periods of high distress.

Soundscape Designer: Suggests "Sound Recipes" based on natural language descriptions (e.g., "high-pitched hiss" -> "Pink Noise with Stream").

Pattern Correlation: Identifies non-obvious links between your sleep, usage, and distress scores.

Clinical Report Translator: Generates a professional summary of your progress for your doctor.

20. SPECTROGRAM & LEVEL METER

Spectrogram: A real-time visualizer of audio frequencies. Use this to verify your **Notch** (look for the "hole" in the graph) or **Acoustic CR** tone alignment.

Audio Level Meter: Displays real-time environmental volume. Habituation is most effective in environments between 45dB and 60dB.

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21. FEEDBACK TOOL

The feedback tool allows users to submit technical bug reports or feature requests. To maintain clinical privacy, you can generate a technical log file to send via email rather than using a persistent database or third-party tracking.

22. SYSTEM VALIDATION ENGINE

High-frequency therapy requires extreme technical precision. This engine audits your hardware and browser settings weekly:

Notch Depth Check: Verifies your filters are achieving at least 40dB of attenuation.

Phase Test: Ensures L/R channels are not inverted.

Sample Rate Check: Confirms your system is not running in low-fidelity "Hands-Free" mode.

23. AUDIO WATCHDOG

Automated Audio Watchdog: Browsers often "suspend" audio in background tabs to save power. This background observer detects stalls and automatically attempts to resume the session, ensuring uninterrupted therapy even when the screen is off.

Audio Status Indicator

Located in the top navigation bar, this provides real-time engine feedback:

- **Audio Active:** The engine is processing sound normally.
- **Audio Suspended:** The browser has paused the audio. Simply interact with the page (click any button) to resume playback.

Technical Appendix

Flexible Audio Sources

Every therapy module supports three input methods:

Internal: Calibrated white, pink, and brown noise.

Custom: User-uploaded relaxing music or nature files.

Live: Capture audio from other browser tabs (Spotify/YouTube) and process it through the Notch or Decorrelation engine in real-time.

Noise Generator (Python-style)

A specialized utility for generating broadband noise with scientific precision, mirroring Python signal-processing behavior. Uses Paul Kellet filter banks for Pink noise and lossy integrators for Brown noise.

Quick Start Summary

Modality	Key Setting	Target Volume	Duration
Notch Therapy	Match Freq / 1.0 Octave	Mixing Point	30-60m Daily
Acoustic CR	Log-spaced Tones	Equal Loudness	60m Daily
Decorrelated Noise	Pink Noise / Wide Stereo	Mixing Point	24/7 Enrichment
Dual-Stimulus	Haptic Sync / Mobile	Comfortable	30m Daily
Broadband Sound	Pink Noise / Nature	Mixing Point	24/7 Enrichment
Binaural Beats	Delta/Theta Offset	Comfortable	Sleep Transitions

24. CLINICAL FAQ

Clinical FAQ

Is this a medical device? No. It is an educational and research tool. It is not intended to diagnose or treat medical conditions.

How loud should the therapy be? Always use the **Mixing Point**. Total masking hides the signal from the brain, preventing it from learning to ignore the tinnitus.

Bluetooth Tip: Ensure your OS is set to "Stereo" output. "Hands-Free AG" mode reduces quality to 8kHz, which will corrupt high-frequency therapy filters.

25. CLINICAL RESOURCES

Professional Summary: A standardized report focusing on audiological metrics (MML, LG, RI) and distress trends (THI) for professional review.

White-Label Integration: Clinics can brand the suite with their own logos on all generated PDF reports.

Remote Monitoring: Clinics can set a custom email address in the user's settings to enable one-clicking exports for remote patient monitoring (supporting CPT 99454).

26. GLOSSARY OF CLINICAL TERMS

Habituation: The neuroplastic process where the brain filters out the tinnitus signal from conscious awareness.

Lateral Inhibition: A physiological process where stimulated neurons suppress the activity of their overactive neighbors (the mechanism behind Notch Therapy).

Minimum Masking Level (MML): The lowest volume level of noise required to just barely hide the tinnitus sound.

Residual Inhibition (RI): The temporary reduction or suppression of tinnitus volume following sound exposure.

Q-factor: A measure of the sharpness of an auditory filter; high Q indicates very tonal tinnitus.

Recruitment: A condition where loudness perception increases abnormally fast, often associated with hearing loss and hyperacusis.

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27. DATA INTERPRETATION GUIDE

Knowing what to look for in your reports is key to measuring progress:

THI Score: Look for a downward trend. A drop of **7 points or more** is clinically significant.

MML Percentage: If you previously required 40% volume to mask and now only need 20%, your brain is successfully "filtering" the signal.

LG Flattening: For hyperacusis sufferers, look for the "Uncomfortable" rating to shift toward higher objective volumes over time.

RI Duration: Increasing silence duration after therapy indicates positive auditory system responsiveness.

28. USER EXPERIENCE TIPS

Home Screen Install: Use "Add to Home Screen" on iOS/Android to install the suite as a PWA. This improves background audio reliability and removes the address bar.

Focus Mode: Enable "Do Not Disturb" during sessions. Incoming notifications can cause the volume to "click" or drop, disrupting the calibrated Mixing Point.

29. AI PRIVACY & DATA SECURITY

The Trahreg Suite is designed with a "Privacy-First" architecture. Understanding how your data is handled is critical, especially when utilizing the AI-powered features (Google Gemini).

1. Clinical Summary Anonymization

When you generate a **Professional Summary** for your clinician, the suite extracts objective metrics such as THI scores, MML levels, and adherence logs. This data is transmitted to the AI model *without* any personally identifiable information (PII) such as your name, email, or device ID. The summary is created as a temporary prompt and is not stored by the application developers.

2. Client-Side Logic

All data synthesis occurs locally on your machine. The "Professional Summary" included in your PDF reports is generated locally within your browser context. No therapy logs or audio data ever leave your device unless you manually export them.

3. Personal API Key Control

The suite requires a user-provided **Google Gemini API Key**. This ensures that you maintain full ownership and control over your interaction history. The developers of this suite have no visibility into the prompts sent to the model or the responses received.

4. Local Data Persistence

All therapeutic progress data is stored in your browser's **Origin-Private Storage** (LocalStorage). It remains strictly local to your device and is never synchronized to a cloud account or third-party server.

Access the suite at: <https://tinnitus.trahreg.com>